

Survival Analysis Methods Correcting for Treatment Switching Effects in RCTs: Theory and SAS/R Code (Part I)

ASA short course recap

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Introduction



In randomized clinical oncology trials, overall survival (OS) is a gold standard for measuring active intervention effectiveness

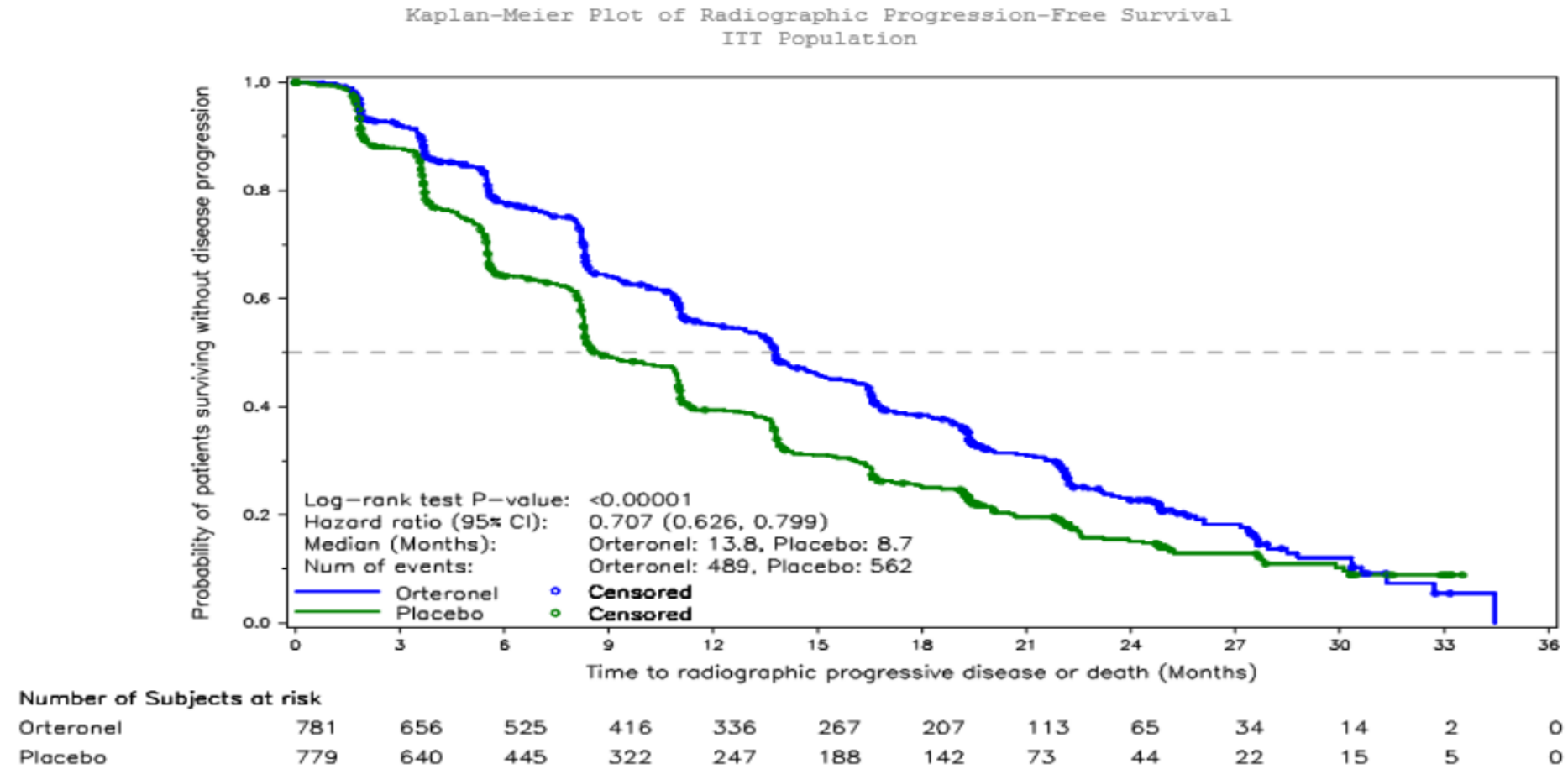
Before a death event is observed, patients may ***switch to alternative treatment***

By trial design, treatment switching may occur

- in the control arm only (i.e., one-way switching)
- in both arms (i.e., two-way switching^[#])

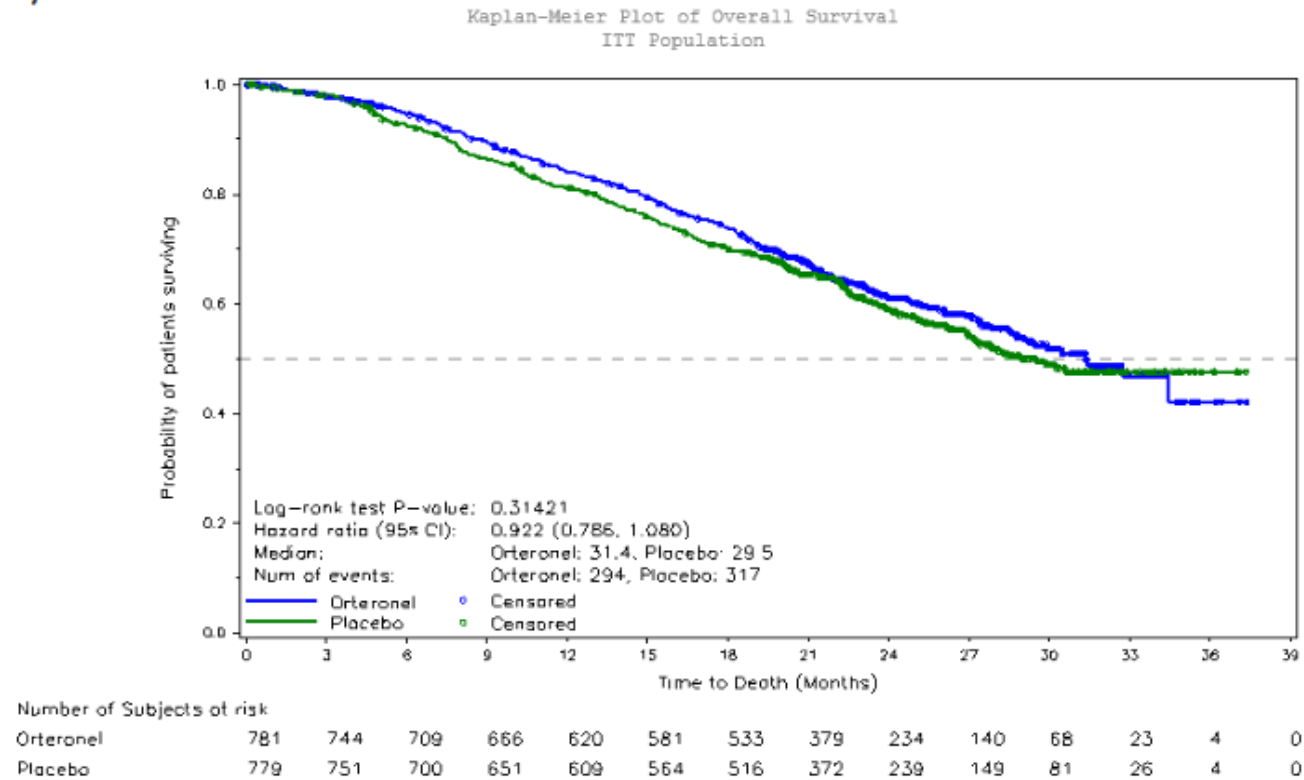
In the presence of one-way or two-way switching, the effect of active intervention on OS may not be directly observable

The RCT showed that Orteronel plus Prednisone improved rPFS compared to placebo plus prednisone, HR = 0.707 (95% CI 0.626-0.799); $p < 0.0001$



However, the ITT analysis did not demonstrate a statistically significant improvement in the study's second primary endpoint of OS. HR for OS = 0.92 (95%CI 0.786-1.080); $p = 0.3142$

Treatment switching may one factor impacting causal inference on the OS benefit. This case study data will be re-analyzed by 3 of the 4 causal inference methods to be discussed in this short course.



Four Causal Inference Methods via Hypothetical Strategy



1

Marginal Structural Model (MSM)

2

Inverse Probability of Censoring Weighting (IPCW)*

3

Rank Preserving Structure Failure Time (RPSFT) Model

4

Two Stage Estimation (TSE)

Marginal Structure Model (MSM)

What Was MSM Originally Designed For?

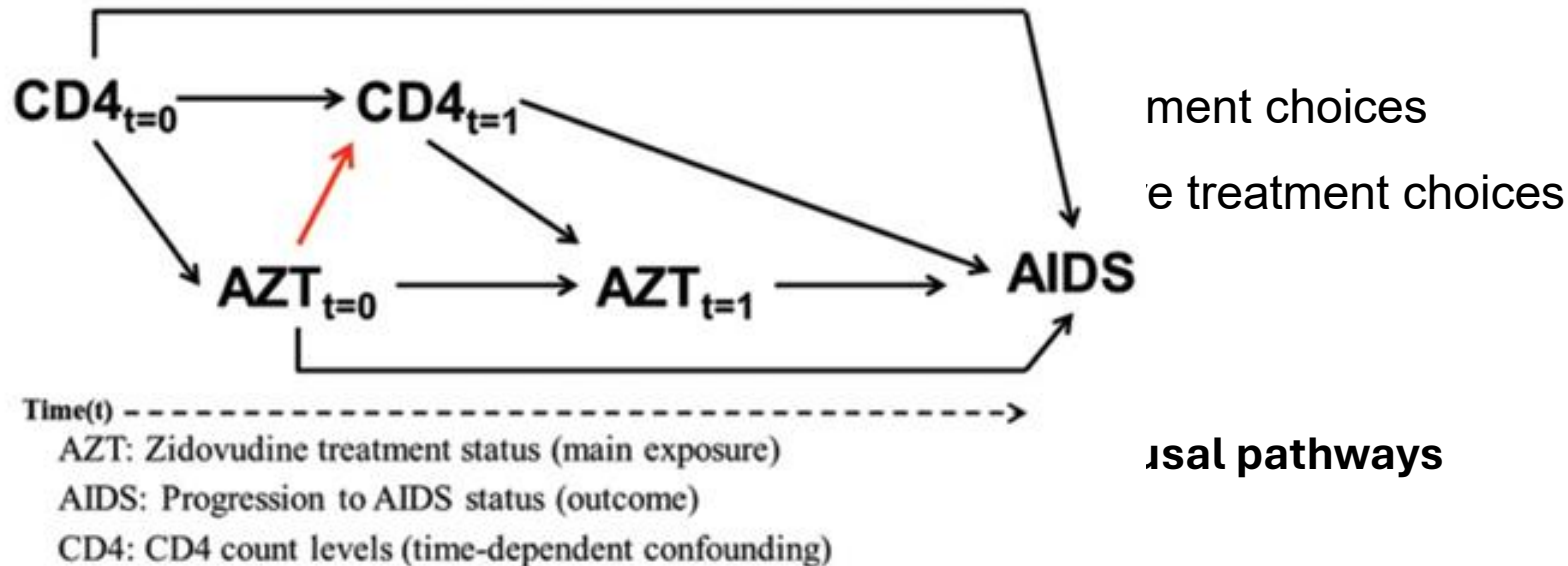
- **MSMs were originally developed for observational studies where treatment changes over time.**
- Clinical status evolves over time (e.g., biomarkers, disease severity, symptoms → time-varying covariates)
- **These time-varying covariates predict survival**
 - Sicker patients are more likely to start or intensify treatment
 - More stable patients may delay or avoid treatment

- **These covariates**

- Effective
- Ineffective

- **Consequences**

Time-varying
→ Adjusting



How MSM Works in observational studies

Conceptually

- MSM Uses **Inverse Probability Weights** (IPW) to Create a Pseudo-Population
- Two major sources of bias:
 - **Time-dependent treatment confounding**
(Health status is influenced by previous treatment and, in turn, affects future treatment decisions and outcomes.)
 - **Informative censoring**
(Worsening clinical status might increase the probability of *treatment discontinuation*, *loss to follow-up*, or other forms of censoring)
- MSM corrects both biases using:
 - **IPTW** (Inverse Probability of Treatment Weight) to make treatment “as-if randomized”
(Treatment no longer depends on evolving health status)
 - **IPCW** (Inverse Probability of Censoring Weight) to make censoring “as-if random”
(Censoring no longer depends on prognosis → removing censoring bias)
- MSM estimates the effects by fitting a model to the weighted Pseudo-Population where confounding has been removed

How MSM Works in observational studies

IPTW & IPCW

- IPTW

$$w = \frac{1}{P(\text{treatment} | \text{current health, prior treatment})}$$

- IPCW

$$w = \frac{1}{P(\text{uncensored at time } t | \text{prior health, prior treatment, prior censor status})}$$

Pooled Logistic Model

- **Stabilized weights** avoid extreme values and keep the MSM model stable.

- $SW_{IPTW} = \frac{P(\text{treatment} | \text{baseline})}{P(\text{treatment} | \text{current health, prior treatment})}$

- $SW_{IPCW} = \frac{P(\text{uncensored} | \text{baseline})}{P(\text{uncensored at time } t | \text{prior health, prior treatment, prior censor status})}$

- Final **MSM weight** = $SW_{IPTW} \times SW_{IPCW}$

How MSM Works in observational studies

Marginal structural Cox model & KM estimates

- After applying the IPTW and IPCW, we fit a **Cox model to the weighted pseudo-population**, where treatment and censoring behave as if randomized.

$$\lambda_T(t|A(t), V) = \lambda_0(t) \exp(\beta_0 V + \beta_1 A(t)) \quad (\text{Hernán et al., 2000})$$

- **A(t)**: treatment status that can change over time
- **V**: baseline covariates
- Because the data are reweighted, **A(t)** can now be treated as *independent of prior health status*.
- $e^{\beta_1} = \text{causal hazard ratio for treatment at time } t$
- Weighted KM-curves can also be calculated as:

$$S(t) = \prod_{t_j \leq t} \left[1 - \frac{\text{weighted deaths at } t_j}{\text{weighted at-risk at } t_j} \right] \quad (\text{Xu et al., 2022})$$

MSM in Observational Studies vs. Treatment-Switching Trials

(Robins et.al., 2000, Xu et.al, 2022)

	MSM in Observational Studies	MSM in Treatment-Switching Trials
Study objective	Estimate the causal effect of a time-varying exposure	Estimate the joint causal effects of the randomized treatment and switching treatment
Type of confounding	Baseline + time-dependent confounders	Time-dependent confounders arising from treatment switching
Randomization	No	Yes (at baseline)
Role of IPW	Breaks the association between treatment and confounders	Breaks the association between switching and confounders
Key challenge	Confounders affected by prior exposure	Time-dependent confounding due to post-progression switching
Ultimate goal	Estimate the causal exposure effect	Recover the true causal effect of the randomized treatment after correcting for switching

MSM in Two-Way Switching Trials (Xu et al., 2022)

- Weight calculation
 - IPCW: same as before → adjusts for censoring bias
 - IPTW: inverse probability of **switching**
- Marginal Structural Cox Model (no interaction)

Fitted on **the weighted pseudo-population** created by IPTW × IPCW

$$\lambda_T(t|R, A(t)) = \lambda_0(t) \exp(\beta_0 V + \beta_1 R + \beta_2 A(t))$$

- R: Randomized treatment arm (baseline)
- A(t): Current treatment status (switching indicator)
- V: Baseline covariates.

Interpretation

- e^{β_1} : HR for active vs. control arm **if no switching had occurred**
- e^{β_2} HR for switching (control → experimental) vs staying on control

MSM in Two-Way Switching Trials (Xu et al., 2022)

Continue

- **When There Is Interaction Between R and A(t)**

$$\lambda_T(t|R, A(t)) = \lambda_0(t) \exp(\beta_0 V + \beta_1 R + \beta_2 A(t) + \beta_3 R \times A(t))$$

- e^{β_1} : HR for active vs. control arm **if no switching had occurred**
- In real trials, after switching occurs, patients may:
 - Discontinue the alternative therapy early
 - Receive **multiple** or **heterogeneous** post-progression treatments
 - Switch to therapies **not fully captured** in the dataset
- → These complexities make the **causal effect of the alternative treatment (β_2) difficult or sometimes impossible** to estimate precisely.
- However, the **randomized treatment effect (β_1) remains valid** if the model and weights are correctly specified.

MSM in Two-Way Switching Trials (Xu et al., 2022)

Reducing Bias

No unmeasured confounding

- Means: all the variables that influence both **treatment decisions** (e.g., switching) **and outcomes** (e.g., survival) are **observed** and **included** in the analysis
- Cannot be directly tested. In RCTs, reduce by **collecting all major prognostic factors** at trial-planning stage.

Positivity (non-zero treatment probability)

- Means: Each patient must have a **positive probability of switching and not switching** for any confounder pattern
- **Structural non-positivity**: e.g., only control-arm patients can switch
- **Random non-positivity**: small N, too many weak covariates, or sparse data late in follow-up.

Marginal Structure Model (Xu et al., 2025)

MSM in one-way switching trials

- In one-way switching designs, only **control-arm** patients can switch to the active treatment after disease progression.
 - Applying IPTW to *all* patients violates **positivity**, because switching probability = 0 in the active arm.
- Solution: Two-step approach
 - Step 1: Apply MSM to control group → removes confounding from switching
 - Compute **IPTW** and **IPCW** as in the two-way switching setup
 - Step 2: Apply MSM to the full data → estimate the **randomized treatment effect**
 - For the control group: using the weights from step1
 - For active-arm patients → set **IPTW = 1** and calculate **IPCW** for active group.
 - e^{β_1} from Step 2 represents the causal effect of randomized active vs. control treatment.

Limitations of MSM in Treatment-Switching Trials

(Xu et al., 2025)

Strong positivity requirement for treatment assignment

If certain covariate patterns *determine* treatment (prob = 0 or 1), IPTW cannot be constructed → bias.

Requires Rich, Fully Observed Time-varying Covariates

Missing or unmeasured prognostic factors → biased causal inference.

Special complications in RCT switching

MSM requires including $R \times A(t)$ interaction by default; curves may cross → median survival hard to interpret.

Inverse Probability of Censoring Weighting (IPCW)

(Robins & Finkelstein, 2000)

- In switching trials, patients may **stop treatment**, **switch treatments**, or become **lost to follow-up**.
- These events are related to patient condition (e.g., progression, toxicity) → **informative (dependent) censoring**.
- Standard KM/Cox/log-rank require **independent censoring**, which is violated here.
- **Key idea:**
 - Treat switching/stop-treatment/loss-to-follow-up as **censoring events** and reweight the data to correct for this dependence.
- **How IPCW Works Conceptually**
 - Patients with low risk of censoring (healthier) → **smaller weights**
 - Patients with high risk of censoring (sicker), but who remain uncensored → **larger weights**
 - $Weights = 1/P(\text{not censored up to } t)$
 - After weighting, the dataset behaves as if **censoring were random**, allowing standard KM/Cox to give unbiased estimates.

IPCW

(Robins & Finkelstein, 2000)

Assumptions

- **No unmeasured confounders for censoring:**

$$C(t) \perp T \mid \{A(s), \quad Z: s \leq t\}$$

- T: True failure time
- C(t): Censoring process at time t (1 = censored/ switched/ stopped; 0 = still observed)
- Z: treatment arm
- **A(t)**: Time-varying clinical covariates up to time t (e.g., CD4, symptoms...)
 - If the changes are captured in **V(t)**, censoring contains **no extra information** about prognosis.
 - Censoring becomes “**explainable**” rather than driven by hidden factors.

- **Why the Assumption Is Needed**

- IPCW upweights patients who remain uncensored despite high risk of censoring, and downweights those with low risk.
- This reweighting only works if **V(t)** captures all factors that influence both censoring and survival.

IPCW (Robins & Finkelstein, 2000)

Weight Construction

1 Estimate the probability of remaining uncensored:
Using pooled logistic regression to compute **the probability of staying uncensored**

2 Compute IPCW = **inverse of the probability of staying uncensored**

3 Derive **stabilized weights** to prevent large variability

After reweighting, the uncensored patients are weighted to represent the full study population that would have been observed if censoring were random, thereby reducing bias from informative censoring

IPCW (Robins & Finkelstein, 2000)

Modeling and Interpretation

- Weighted KM and log-rank test:

$$S(t) = \prod_{t_j \leq t} \left[1 - \frac{\text{weighted deaths at } t_j}{\text{weighted at-risk at } t_j} \right]$$

- Weighted Cox $\lambda_T(t|\bar{A}(t), V, R) = \lambda_0(t)e^{\{\beta_0 V + \beta_1 R\}}$.

Where

- ✓ R is randomized treatment arm.
 - ✓ V is a baseline covariate vector.
 - ✓ A(t) is a treatment arm switching indicator; overbar indicates history up to time t.
- Note:
 - Data are censored at time of treatment switching, therefore,
 - The IPCW model contains no A(t) term, nor A(t) by R interaction term
 - The above IPCW Cox model may cover either one-way or two-way switching
 - If V values are balanced between treatment arms, V may be dropped from the IPCW model.

IPCW

Key takeaways

Accommodating

Accommodating the “No Unmeasured Confounders” Assumption

- In RCTs this assumption is not testable.
- It may not be perfectly true, but can be made approximately valid through careful trial design and thorough collection of prognostic factors.

Avoiding

Avoiding Positivity Violations

- Structural positivity violation: If all patients switch → IPCW cannot be applied.
- High switching proportion (>70%) → IPCW cannot be applied.
- Very low switching proportion → IPCW cannot be applied.

Ensuring

Ensuring Correct Model Specification

- Follow the modeling guidelines recommended for MSM.
- With stabilized weights, do not omit baseline confounders in the extended Cox model if they are imbalanced between randomized groups.

Summary of MSM vs IPCW for Treatment-Switching Trials

(Xu et al., 2022)

	MSM	IPCW
Goal	Recover causal effect as if no switching , using weights	Same goal; correct bias from informative censoring
How It Works	IPTW × IPCW; reweights treatment and censoring	Treat switching as censoring ; weight for follow-up
Treatment Switching	Keep post-switch data	Censor at switching → drop later data
Estimand	Effect of randomized arm and alternative treatment	Effect of staying on randomized arm
Efficiency	Higher (more data retained)	Lower (information loss)
Strengths	Lower bias; higher power; better type I error (when correctly specified)	Simple; intuitive; good when switching patterns are simple
Shared Limitations	Require rich time-varying covariates; sensitive to non-positivity; need correct weight models; extreme weights need truncation	
MSM-Specific Limits	Must model treatment + censoring; requires R × A(t) interaction	—
IPCW-Specific Limits	—	Information loss → larger SE; cannot estimate effect of alternative treatment; tends to underestimate effect

Reference

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